William Merrill

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Research Interests

Main Threads

1. Expressivity and inductive biases of neural sequence models for implementing algorithms, representing the syntax and semantics of natural language, and solving reasoning problems
2. The theory of learning semantic structure from raw text corpora

Broader Interests The following theoretical frameworks and their applications to analyzing language models, solving NLP problems, and understanding human language:

- Formal languages and automata
- Computational complexity; especially circuits
- Formal semantics

Experience

Allen Institute for AI	2024–	**Research Intern** (part-time, 20%), AllenNLP
Allen Institute for AI	2023	**Research Intern**, AllenNLP
Google Research	2022	**Student Researcher**, Speech & Language Algorithms
New York University	2021–	**Ph.D. Student**, Center for Data Science
Allen Institute for AI	2019–21	**PYI** (predoctoral researcher), AllenNLP
Yale University	2015–19	**B.Sc.** with distinction in Computer Science
		B.A. with distinction in Linguistics (*cum laude*)
		Note of excellence on thesis: *Sequential neural networks as automata*
Google	2018	**Software Engineering Intern**
Boston College	2017	**Research Intern**, Language Learning Lab
New York University	2013–15	**Research Intern**, Morphology Lab

Academic Group Affiliations

CapLab & **ML**^{2}, NYU	*Tal Linzen*	2021–
AllenNLP, Ai2	*Noah A. Smith, Yoav Goldberg, Roy Schwartz*	2019–21
CLAY, Yale	*Robert Frank, Dana Angluin*	2016–19
L^3, Boston College	*Joshua Hartshorne, Sven Dietz*	2017
MorphLab, NYU	*Alec Marantz, Phoebe Gaston*	2013–15

Archival Publications

[1] **W. Merrill**, N. A. Smith, and Y. Elazar. Evaluating n-Gram Novelty of Language Models Using Rusty-DAWG. *EMNLP*. Nov. 2024.

[2] J. Pfau, **W. Merrill**, and S. Bowman. Lets Think Dot by Dot: Hidden computation in transformer language models. *COLM*. Oct. 2024.

[3] D. Groeneveld, I. Beltagy, P. Walsh, A. Bhagia, R. Kinney, O. Tafjord, A. H. Jha, H. Ivison, I. Magnusson, Y. Wang, S. Arora, D. Atkinson, R. Authur, K. R. Chandu, A. Cohan, J. Dumas, Y. Elazar, Y. Gu, J. Hessel, T. Khot, **W. Merrill**, J. Morrison, N. Muennighoff, A. Naik, C. Nam, M. E. Peters, V. Pyatkin, A. Ravichander, D. Schwenk, S. Shah, W. Smith, E. Strubell, N. Subramani, M. Wortsman, P. Dasigi, N. Lambert, K. Richardson, L. Zettlemoyer, J. Dodge, K. Lo, L. Soldaini, N. A. Smith, and H. Hajishirzi. OLMo: Accelerating the Science of Language Models. *ACL*. Aug. 2024.

[4] **W. Merrill**, Z. Wu, N. Naka, Y. Kim, and T. Linzen. Can You Learn Semantics Through Next-Word Prediction? The Case of Entailment. *ACL*. Aug. 2024.

[5] **W. Merrill**, J. Petty, and A. Sabharwal. The Illusion of State in State-Space Models. *ICML*. July 2024.

[6] M. Zhang, O. Press, **W. Merrill**, A. Liu, and N. A. Smith. How Language Model Hallucinations Can Snowball. *ICML*. July 2024.

[7] **W. Merrill** and A. Sabharwal. The Expressive Power of Transformers with Chain of Thought. *ICLR*. May 2024.

[8] L. Strobl, **W. Merrill**, G. Weiss, D. Chiang, and D. Angluin. What Formal Languages Can Transformers Express? A Survey. *TACL* (May 2024).

[9] **W. Merrill** and A. Sabharwal. A Logic for Expressing Log-Precision Transformers. *NeurIPS*. Dec. 2023.

[10] **W. Merrill**. Formal Languages and the NLP Black Box. *Developments in Language Theory*. Ed. by F. Drewes and M. Volkov. Cham: Springer Nature Switzerland, June 2023.

[11] **W. Merrill** and A. Sabharwal. The Parallelism Tradeoff: Limitations of Log-Precision Transformers. *TACL* (June 2023).

[12] **W. Merrill**, N. Tsilivis, and A. Shukla. A Tale of Two Circuits: Grokking as Competition of Sparse and Dense Subnetworks. *ICLR Workshop on Mathematical and Empirical Understanding of Foundation Models*. May 2023.

[13] Z. Wu, **W. Merrill**, H. Peng, I. Beltagy, and N. A. Smith. Transparency Helps Reveal When Language Models Learn Meaning. *TACL* (2023).

[14] **W. Merrill**, A. Warstadt, and T. Linzen. Entailment Semantics Can Be Extracted from an Ideal Language Model. *CoNLL*. Abu Dhabi, United Arab Emirates (Hybrid), Dec. 2022.

[15] **W. Merrill**, A. Sabharwal, and N. A. Smith. Saturated Transformers are Constant-Depth Threshold Circuits. *TACL* (Aug. 2022).

[16] S. Subramanian, **W. Merrill**, T. Darrell, M. Gardner, S. Singh, and A. Rohrbach. Re-CLIP: A Strong Zero-Shot Baseline for Referring Expression Comprehension. *ACL*. Dublin, Ireland, May 2022.

[17] M. Gardner, **W. Merrill**, J. Dodge, M. Peters, A. Ross, S. Singh, and N. A. Smith. Competency Problems: On Finding and Removing Artifacts in Language Data. *EMNLP*. Online and Punta Cana, Dominican Republic, Nov. 2021.

[18] **W. Merrill**, V. Ramanujan, Y. Goldberg, R. Schwartz, and N. A. Smith. Effects of Parameter Norm Growth During Transformer Training: Inductive Bias from Gradient Descent. *EMNLP*. Online and Punta Cana, Dominican Republic, Nov. 2021.

[19] **W. Merrill**, Y. Goldberg, R. Schwartz, and N. A. Smith. Provable Limitations of Acquiring Meaning from Ungrounded Form: What Will Future Language Models Understand? *TACL* (Sept. 2021).

[20] **W. Merrill**, G. Weiss, Y. Goldberg, R. Schwartz, N. A. Smith, and E. Yahav. A Formal Hierarchy of RNN Architectures. *ACL*. Online, July 2020.

[21] L. L. Wang, K. Lo, Y. Chandrasekhar, R. Reas, J. Yang, D. Burdick, D. Eide, K. Funk, Y. Katsis, R. M. Kinney, Y. Li, Z. Liu, **W. Merrill**, P. Mooney, D. A. Murdick, D. Rishi, J. Sheehan, Z. Shen, B. Stilson, A. D. Wade, K. Wang, N. X. R. Wang, C. Wilhelm, B. Xie, D. M. Raymond, D. S. Weld, O. Etzioni, and S. Kohlmeier. CORD-19: The COVID-19 Open Research Dataset. *ACL Workshop on NLP for COVID-19*. Online, July 2020.

[22] **W. Merrill**. Sequential Neural Networks as Automata. *ACL Workshop on Deep Learning and Formal Languages*. Florence, Aug. 2019.

[23] **W. Merrill**, L. Khazan, N. Amsel, Y. Hao, S. Mendelsohn, and R. Frank. Finding Hierarchical Structure in Neural Stacks Using Unsupervised Parsing. *ACL Workshop BlackboxNLP*. Florence, Italy, Aug. 2019.

[24] **W. Merrill**, G. Stark, and R. Frank. Detecting Syntactic Change Using a Neural Part-of-Speech Tagger. *ACL Workshop on Computational Approaches to Historical Language Change*. Florence, Italy, Aug. 2019.

[25] Y. Hao, **W. Merrill**, D. Angluin, R. Frank, N. Amsel, A. Benz, and S. Mendelsohn. Context-Free Transductions with Neural Stacks. English. *EMNLP Workshop BlackboxNLP*. Brussels, Belgium, Nov. 2018.

[26] J. Kasai, R. Frank, P. Xu, **W. Merrill**, and O. Rambow. End-to-End Graph-Based TAG Parsing with Neural Networks. *NAACL*. 2018.

Non-Archival Publications

[27] **W. Merrill** and N. Tsilivis. *Extracting Finite Automata from RNNs Using State Merging*. Jan. 2022.

[28] **W. Merrill**. *Formal Language Theory Meets Modern NLP*. Feb. 2021.

[29] **W. Merrill**. *On the Linguistic Capacity of Real-Time Counter Automata*. Sept. 2020.

[30] **W. Merrill**. A Semantics of Subordinate Clauses Using Delayed Evaluation. *Toronto Undergraduate Linguistics Conference* (2018).

Press Coverage

[1] **Quanta Magazine**. How Chain-of-Thought Reasoning Helps Neural Networks Compute. March 2024.

[2] **NYU CDS Blog**. Language Models Provide Insight into Linguistic Redundancy. March 2024.

[3] **Washington Post**. Honestly, I Love When AI Hallucinates. Dec. 2023.

[4] **NYU CDS Blog**. The Logic of Transformers: William Merrill's Step Towards Understanding Large Language Models' Limits and Hallucinations. Oct 2023.

[5] **NYU CDS Blog**. Can Language Models Learn Meaning Just By Observing Text?. Oct 2022.

Talks

The Parallelism Tradeoff: Understanding Transformer Expressivity Through Circuit Complexity

[1] TAMI Lab, UMass Amherst, 2024

[2] Aspen Physics Institute Workshop on Foundation Models, Aspen, 2024

[3] Transformers as a Computational Model Workshop, Simons Institute, Berkeley, 2024

[4] NYC GenAI Collective, 2024

[5] Guest Lecture in *Natural Language Understanding* Course, NYU, 2024

[6] Fellowship Finalist Reception, Two Sigma, 2024

[7] Transformer Theory Seminar, Flatiron Institute, 2023

[8] Limitations of LMs Workshop, Bielefeld University, 2023

[9] Lingo Group, MIT CSAIL, 2023

[10] CDS Depth Qualifying Exam, NYU, 2023

[11] Microsoft Research NYC, 2022

Formal Languages and Neural Models for Learning on Sequences

[12] Angluin Invited Tutorial, ICGI, 2023

Formal Languages and the NLP Black Box

[13] Invited Talk, Developments in Language Theory, 2023

The Expressive Power of Transformers with Chain of Thought

[14] FLaNN Discord, 2024

The Illusion of State in State-Space Models

[15] NLP Seminar, UMass Amherst, 2024

[16] Speech and Language Algorithms, Google Research, 2024

[17] FLaNN Discord, 2024

Saturated Transformers are Constant-Depth Threshold Circuits

[18] TACL Track, EMNLP, 2022

[19] FLaNN Discord, 2022

[20] Foundations of Language Processing Group, Umeå University, 2022

[21] ML for Code Seminar, MILA, 2022

Can You Learn Semantics Through Next-Word Prediction? The Case of Entailment

[22] LUNAR Lab, Brown University, 2024

[23] Transformer Theory Seminar, Flatiron Institute, 2024

Entailment Semantics Can Be Extracted from an Ideal Language Model

[24] Linguae Seminar, Institut Jean Nicod, 2023

[25] Invited Speaker, NYC Philosophy of Language Workshop, 2023

[26] Guest Lecture in *Computational Linguistics & Cognitive Science*, NYU, 2023

[27] CoNLL Workshop, EMNLP, 2022

[28] Foundations of Language Processing Group, Umeå University, 2022

[29] Journal Club, ArthurAI, 2022

[30] CompLang Seminar, MIT, 2022

[31] Semantics Seminar, NYU, 2022

Neural Networks as Automata

[32] Speech and Language Algorithms, Google Research, 2022

Benchmarking Whether LMs Copy from Their Pretraining Data

[33] AllenNLP Team Meeting, Ai2, 2023

Competency Problems: On Finding and Removing Artifacts in Language Data

[34] Journal Club, ArthurAI, 2021

[35] ML Track, EMNLP, 2021

Parameter Norm Growth During Transformer Training: Inductive Bias from Gradient Descent

[36] ML Track, EMNLP, 2021

Provable Limitations of Acquiring Meaning from Ungrounded Form: What Will Future Language Models Understand?

[37] All Hands, Ai2, 2021

[38] Noah's ARK, UW, 2020

Context-Free Transductions with Neural Stacks

[39] Blackbox NLP, EMNLP, 2018

Invited Workshop and Seminar Participation

[1] Dagstuhl Seminar 25282 on *Theory of Neural Language Models*, July 2025

[2] Dagstuhl Seminar 25061 on *Logic and Neural Networks*, Feb 2025

[3] Aspen Meeting on Foundation Models, Oct 2024

[4] Simons Institute workshop on *Transformers as a Computational Model*, Sept 2024

[5] Bielefeld University workshop on *Limitations of Large Language Models*, Nov 2023

[6] ICGI invited tutorial, July 2023

[7] Developments in Language Theory invited lecture, June 2023

Poster Presentations

[1] **ACL**, CMCL Workshop, 2024
Can You Learn Semantics Through Next-Word Prediction? The Case of Entailment

[2] **ACL**, Findings Track, 2024
Can You Learn Semantics Through Next-Word Prediction? The Case of Entailment

[3] **ICML**, DMLR workshop, 2024
Evaluating n-Gram Novelty of Language Models Using Rusty-DAWG

[4] **ICML**, 2024
How Language Model Hallucinations Can Snowball

[5] **ICML**, 2024
The Illusion of State in State-Space Models

[6] **ICLR**, 2024
The Expressive Power of Transformers with Chain of Thought

[7] **NeurIPS**, M3L Workshop, 2024
The Expressive Power of Transformers with Chain of Thought

[8] **NeurIPS**, 2024
A Logic for Expressing Log-Precision Transformers

[9] **Philosophy of Deep Learning Workshop**, NYU, 2023
Entailment Semantics Can Be Extracted from an Ideal Language Model

[10] **EMNLP**, ML Track, 2021
Effects of Parameter Norm Growth During Transformer Training: Inductive Bias from Gradient Descent

[11] **EMNLP**, ML Track, 2021
Provable Limitations of Acquiring Meaning from Ungrounded Form: What Will Future Language Models Understand?

[12] **ACL**, Deep Learning and Formal Languages, 2019
Sequential Neural Networks as Automata

[13] **ACL**, Blackbox NLP, 2019
Finding Hierarchical Structure in Neural Stacks Using Unsupervised Parsing

Teaching Experience

NYU

[1] **Lead TA** for *Natural Language Processing*, Tal Linzen (NYU, Fall 2022)

Yale

[2] **TA** for *Artificial Intelligence*, Dragomir Radev (Yale, Spring 2019)

[3] **TA** for *Natural Language Processing*, Dragomir Radev (Yale, Fall 2018)

[4] **TA** for *Artificial Intelligence*, Dragomir Radev (Yale, Spring 2017)

Broader Community

[5] **TA** for *Introductory NLP* at NYU AI School (Summer 2024)

[6] **TA** for *Introductory NLP* at NYU AI School (Spring 2022)

[7] **Instructor** and **guest lecturer** for CodeHaven (2016-2018)

[8] **Instructor** for Splash at Yale: *Viking Runes*, *The Politics of Skyrim*, *DECLASSIFIED: The History of Codebreaking* (2016-2017)

Service

Reviewing

EMNLP Demo Track	Sept 2024	*Conference*	5 reviews
JMLR	Aug 2024	*Journal*	1 review
ARR	June 2024	*Conference*	3 reviews
NGSM	May 2024	*Workshop*	4 reviews
COLM	May 2024	*Conference*	3 reviews
ICLR	Oct 2023	*Conference*	3 reviews (+1 emergency)
M3L	Oct 2023	*Workshop*	3 reviews
GenBench	Sept 2023	*Workshop*	3 reviews
NeurIPS	July 2023	*Conference*	1 emergency review
JMLR	June 2023	*Journal*	1 review
ACL SRW	May 2023	*Workshop*	2 reviews
ICGI	April 2023	*Conference*	2 reviews
ACL	Feb 2023	*Conference*	1 review
Proc. of Royal Society A	Jan 2023	*Journal*	1 review
ARR	Nov 2022	*Conference*	1 review
Inverse Scaling Prize	Sept 2022	*Competition*	7 reviews
TheoretiCS	July 2022	*Journal*	1 review
ARR	April 2022	*Conference*	1 review
ARR	Jan 2022	*Conference*	2 review
ARR	Dec 2021	*Conference*	3 reviews
ARR	Nov 2021	*Conference*	1 review
CL	2021	*Journal*	1 review
ACL	2021	*Conference*	6 reviews
EACL	2021	*Conference*	4 reviews
EMNLP	2020	*Conference*	2 reviews
Neural Networks	2020	*Journal*	1 review

Session Chairing

ICGI	July 2023
DLT	June 2023

Other

[1] **NYC AI School**, organizer (2024)

[2] **ML2 Seminar**, organizer (2024)

[3] **CAP Lab Website**, maintainer (2023)

[4] **FLaNN Discord**, moderator, scheduled and hosted talks (2022)

[5] **NYC AI School**, volunteer instructor (2022)

[6] **AllenNLP Hackathon**, technical support (2021)

[7] **AllenNLP Tutorial**, chapter author (2020)

[8] **Yale Tangut Language Workshop**, videographer and technical support (2018)

[9] **Yale Kitan Language Workshop**, videographer and technical support (2016)

[10] **CodeHaven**, student volunteer (2016–18)

[11] **Splash at Yale**, volunteer instructor (2016–17)

Selected Public Software

[1] **Rusty-DAWG**: Efficient data structures to search massive pretraining corpora in constant time (lead developer, Rust)

[2] **AllenNLP**: Open-source NLP framework (contributor, Python)

[3] **The Book of Thoth**: Puzzle game with compositional spell casting in Middle Egyptian hieroglyphs (lead developer, Java)

[4] **DraftNet**: Dota 2 drafting using neural networks (lead developer, Python)

[5] **StackNN**: Differentiable stacks, queues, and dequeues in PyTorch (lead developer, Python)

Blog Posts

Research Content

[1] *A Formal Hierarchy of RNN Architectures* (2020)

[2] *Theory of Saturated Neural Networks* (2019)

[3] *The State of Interpretability in NLP* (2019, outdated!)

[4] *Word2vec Analysis of the Voynich Manuscript* (2018)

[5] *Review: Learning to Transduce with Unbounded Memory* (2018)

[6] *Capsule Networks for NLP* (2018)

Literary Translations

[7] *The Wanderer* (Old English → English)

[8] *After Ragnarok* (Old Norse → English)

[9] *The Saga of Mary* (Old Norse → English)

Awards and Grants

[1] **Two Sigma PhD Fellowship** (2024)

[2] First annual **Angluin Invited Tutorial Speaker** (ICGI 2023)

[3] **NSF Graduate Student Research Fellowship** (2022)

[4] **Student Travel Grant** from Naver Labs to attend DELFOL workshop at ACL (2019)

[5] **Mellon Grant** for senior thesis from Benjamin Franklin College at Yale University (2019)

[6] **Grace Hopper Prize** (computer science project award) finalist (2017)

[7] Yale College **freshman rap battle champion** (2016)

[8] **Rising Scientist Award** presented by the Child Mind Institute (2015)

[9] **Study of American History Award** from the Society of Mayflower Descendants (2013)

[10] National Latin Exam ***cum honore maximo egregio*** (2010)

Selected Coursework

Theoretical Computer Science and Formal Languages

[1] *Inference and Representation* (NYU, 2022)

[2] *Foundations of Machine Learning* (NYU, 2022)

[3] *Computational Complexity Theory* (Yale, 2018)

[4] *Computability and Logic* (Yale, 2017)

[5] *Design and Analysis of Algorithms* (Yale, 2017)

[6] *Computing Meanings* (Yale, 2016)

[7] *Introduction to Computer Science* (Yale, 2015)

[8] *Formal Foundations of Linguistic Theory* (Yale, 2015)

Deep Learning and Natural Language Processing

[9] *Seminar: Scaling Laws, the Bitter Lesson, and AI Research* (NYU, 2021)

[10] *Ph.D. Introduction to Data Science* (NYU, 2021)

[11] *Seminar: Selected Topics in Neural Networks* (Yale, 2019)

[12] *Seminar: Advanced Natural Language Processing* (Yale, 2018)

[13] *Computational Vision and Biological Perception* (Yale, 2018)

[14] *Neural Networks and Language* (Yale, 2018)

[15] *Deep Learning Theory and Applications* (Yale, 2018)

[16] *Natural Language Processing* (Yale, 2017)

Other Linguistics

[17] *Hybrid Grammars: Language Contact and Change* (Yale, 2019)

[18] *Phonology I* (Yale, 2018)

[19] *The Voynich Manuscript* (Yale, 2018)

[20] *Indo-European Linguistics* (Yale, 2018)

[21] *Syntax I* (Yale, 2017)

[22] *Seminar: Beowulf and the Northern Heroic Tradition* (Yale, 2017)

[23] *Medieval Latin Paleography* (Yale, 2016)

[24] *Semantics I* (Yale, 2016)

[25] *Old English* (Yale, 2015)

Other Computer Science

[26] *Big Data* (NYU, 2022)

[27] *Systems Programming Techniques and Computer Organization* (Yale, 2017)

[28] *Data Structures and Programming Techniques* (Yale, 2016)

Continuous Math

[29] *Introduction to Analysis* (Yale, 2017)

[30] *MATH 231: Vector Calculus and Linear Algebra II* (Yale, 2016)

[31] *MATH 230: Vector Calculus and Linear Algebra I* (Yale, 2015)

Reading Groups

[31] *Nonlinear Dynamical Systems* (Ai2, 2021)

[32] *Deep Learning Theory* (Ai2, 2020)

Languages

[1] **Modern:** English (Native), Icelandic (Intermediate), German (B1)

[2] **Ancient:** Latin, Old Norse, Old English

[3] **Coding:** Python, Java, C, Rust, Haskell, deep learning libraries, *inter alias*